

Quiz 2

Setup

```
In [11]: # load datasets
im <- read.csv("InfantMortality.csv")
sat <- read.csv("SATGPA.csv")
# check variable names
names(im); names(sat)
head(im)
```

'Mortality' · 'Year'

'MathSAT' · 'VerbalSAT' · 'GPA'

A data.frame: 6 × 2

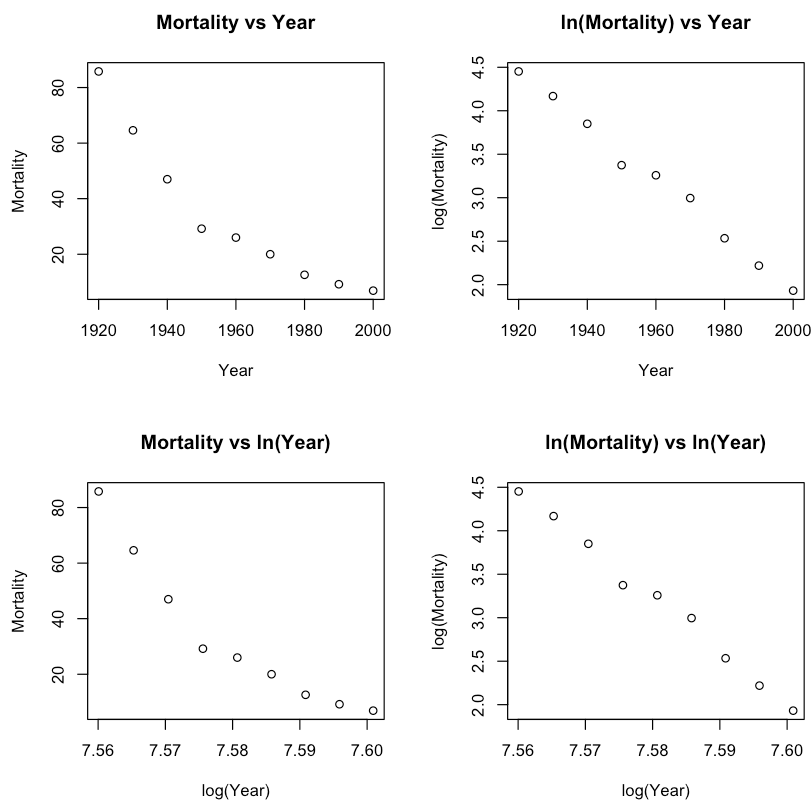
	Mortality	Year
	<dbl>	<int>
1	85.8	1920
2	64.6	1930
3	47.0	1940
4	29.2	1950
5	26.0	1960
6	20.0	1970

1(a)

```
In [13]: # plot and fit raw, ln and log-log versions,
# compare R^2 value (which best linearizes the data)
par(mfrow = c(2, 2))
plot(Mortality ~ Year, data = im, main = "Mortality vs Year")
plot(log(Mortality) ~ Year, data = im, main = "ln(Mortality) vs Year")
plot(Mortality ~ log(Year), data = im, main = "Mortality vs ln(Year)")
plot(log(Mortality) ~ log(Year), data = im, main = "ln(Mortality) vs ln(Year)")
par(mfrow = c(1, 1))

forms <- list(Mortality ~ Year, log(Mortality) ~ Year,
              Mortality ~ log(Year), log(Mortality) ~ log(Year))
for (f in forms) {
  cat(deparse(f), " R^2 =", round(summary(lm(f, data = im))$r.squared, 4),
  }
```

```
Mortality ~ Year    R^2 = 0.8888
log(Mortality) ~ Year    R^2 = 0.9935
Mortality ~ log(Year)    R^2 = 0.8922
log(Mortality) ~ log(Year)    R^2 = 0.9934
```



1

```
In [15]: # fitted line and LSE with residual plot to check fit
m1 <- lm(log(Mortality) ~ Year, data = im)
summary(m1)

b <- coef(m1)
cat(sprintf("ln(Mortality)-hat = %.4f + (%.5f) * Year\n", b[1], b[2]))

plot(fitted(m1), resid(m1), xlab = "Fitted", ylab = "Residuals")
abline(h = 0)
```

Call:

```
lm(formula = log(Mortality) ~ Year, data = im)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.139593	-0.031867	-0.003877	0.023107	0.113316

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	65.0699354	1.8910455	34.41	4.54e-09 ***
Year	-0.0315673	0.0009647	-32.72	6.44e-09 ***

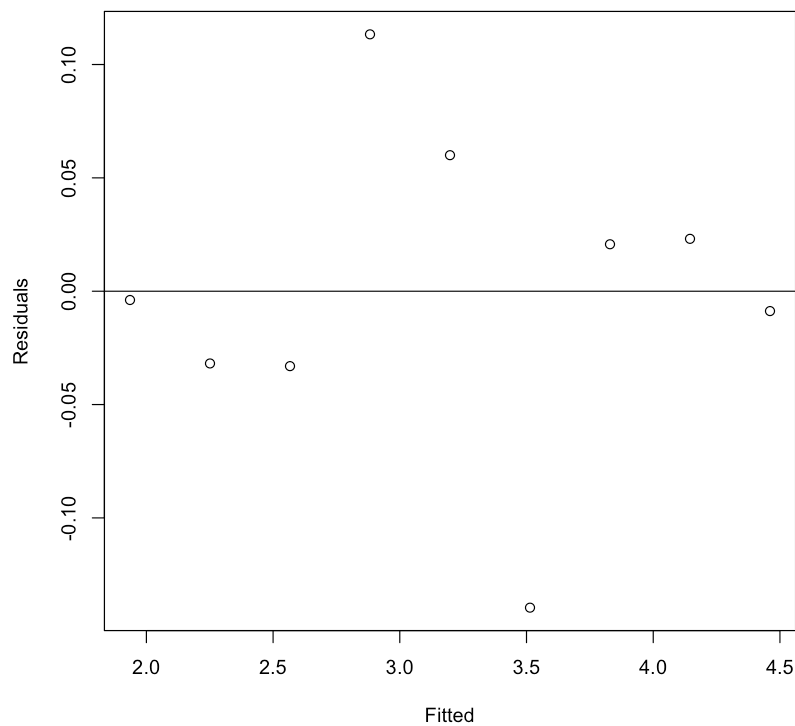
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07473 on 7 degrees of freedom

Multiple R-squared: 0.9935, Adjusted R-squared: 0.9926

F-statistic: 1071 on 1 and 7 DF, p-value: 6.444e-09

$$\ln(\text{Mortality})\text{-hat} = 65.0699 + (-0.03157) * \text{Year}$$



1(b)

```
In [16]: # prediction for 2009
pred_ln <- predict(m1, data.frame(Year = 2009))
pred_ln
exp(pred_ln) # back-transform: deaths per 100,000 infants
```

1: 1.651292754309

1: 5.21371552565186

Problem 2

```
In [17]: # runs t-test for correlation and confirms it with the slope test to get:
# r, t statistic, and p-value for the conclusion
cor.test(sat$VerbalSAT, sat$GPA) # r, t, df, P-value

m2 <- lm(GPA ~ VerbalSAT, data = sat)
summary(m2) # same t and P for the slope

plot(GPA ~ VerbalSAT, data = sat)
abline(m2)
```

Pearson's product-moment correlation

```

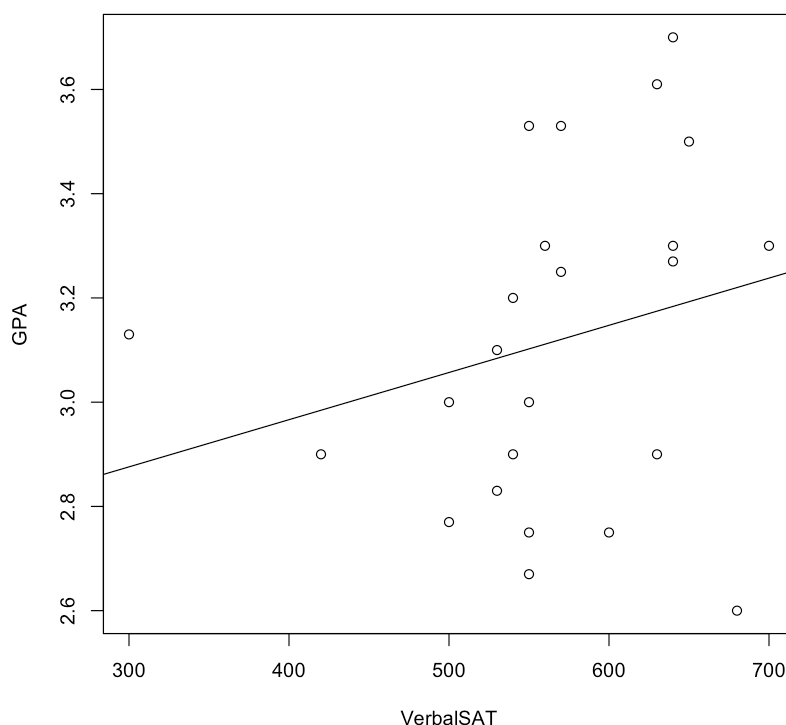
data: sat$VerbalSAT and sat$GPA
t = 1.1825, df = 22, p-value = 0.2496
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.1763308  0.5896995
sample estimates:
      cor
0.2444543
Call:
lm(formula = GPA ~ VerbalSAT, data = sat)

Residuals:
    Min       1Q   Median       3Q      Max
-0.62002 -0.25932  0.03885  0.20502  0.51621

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.6042036   0.4377919   5.948  5.5e-06 ***
VerbalSAT    0.0009056   0.0007659   1.182   0.25
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3154 on 22 degrees of freedom
Multiple R-squared:  0.05976,    Adjusted R-squared:  0.01702
F-statistic: 1.398 on 1 and 22 DF,  p-value: 0.2496

```



```

In [6]: cor.test(sat$VerbalSAT, sat$GPA)      # r, t, df, P-value

m2 <- lm(GPA ~ VerbalSAT, data = sat)

```

```
summary(m2) # same t and P for the slope

plot(GPA ~ VerbalSAT, data = sat)
abline(m2)
```

Pearson's product-moment correlation

```
data: sat$VerbalSAT and sat$GPA
t = 1.1825, df = 22, p-value = 0.2496
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
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```

Residuals:

Min	1Q	Median	3Q	Max
-0.62002	-0.25932	0.03885	0.20502	0.51621

Coefficients:

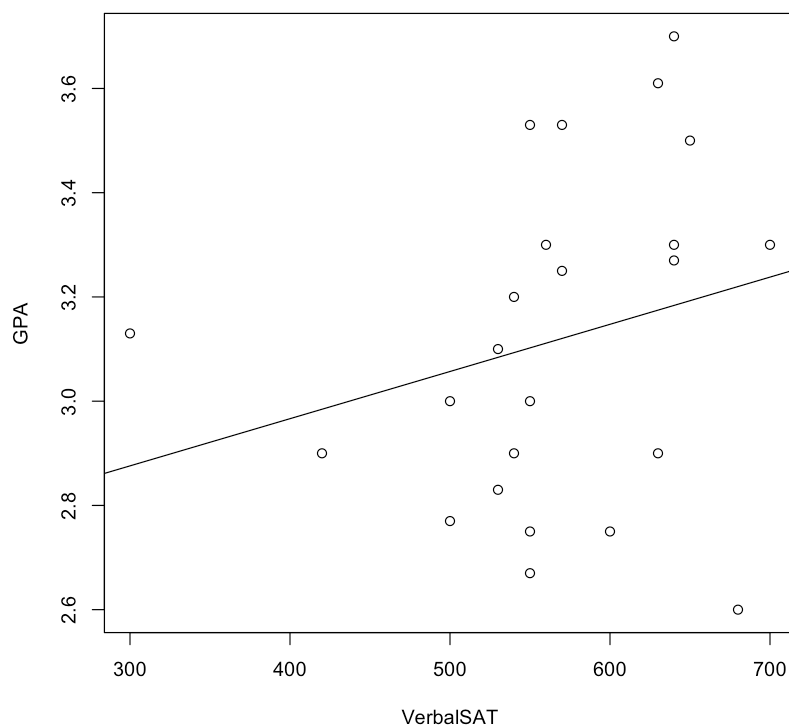
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.6042036	0.4377919	5.948	5.5e-06 ***
VerbalSAT	0.0009056	0.0007659	1.182	0.25

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3154 on 22 degrees of freedom

Multiple R-squared: 0.05976, Adjusted R-squared: 0.01702

F-statistic: 1.398 on 1 and 22 DF, p-value: 0.2496



3(b)

```
In [18]: # builds interval for population slope of elevation
# df = 70
tstar <- qt(0.95, df = 72 - 2)
tstar
1.9195 + c(-1, 1) * tstar * 0.6253 # approx (0.877, 2.962)
```

1.66691447905596

0.877178376246311 · 2.96182162375369

Bonus

```
In [20]: # numerically check that
# slope t-test and correlation t-test give the same statistic
summary(m2)$coefficients["VerbalSAT", "t value"] # slope t
cor.test(sat$VerbalSAT, sat$GPA)$statistic # correlation t (should n

# Problem 3 sanity check: r from R^2 = 0.119, n = 72
r3 <- sqrt(0.119)
r3 * sqrt(70) / sqrt(1 - r3^2) # about 3.07
```

1.18246718303219

t: 1.1824671830322

3.07492513497452